

BiScale-GTR: Fragment-Aware Graph Transformers for Multi-Scale Molecular Representation Learning

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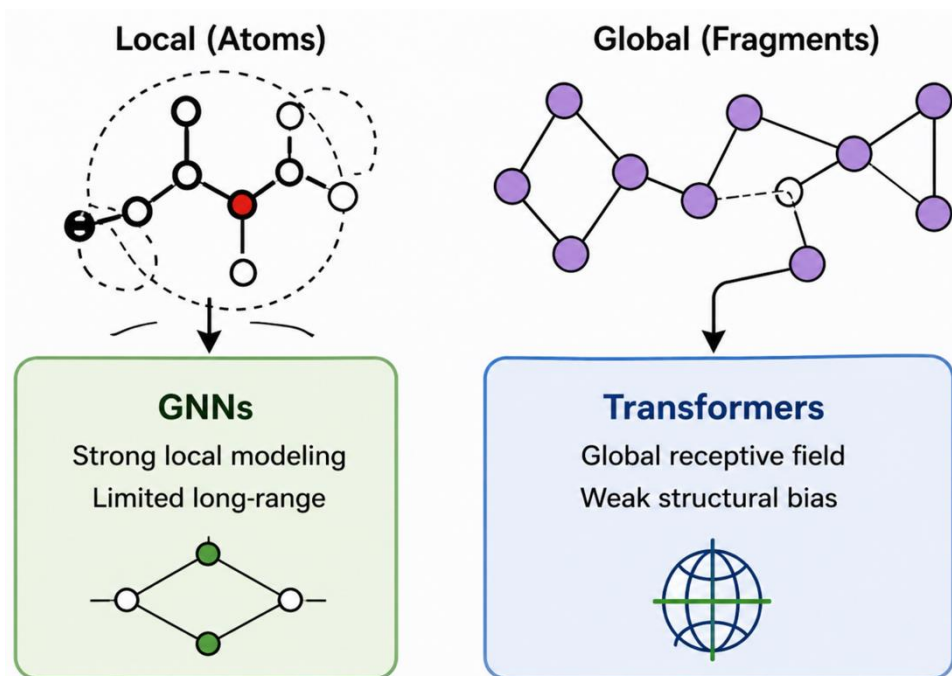


1. Motivation

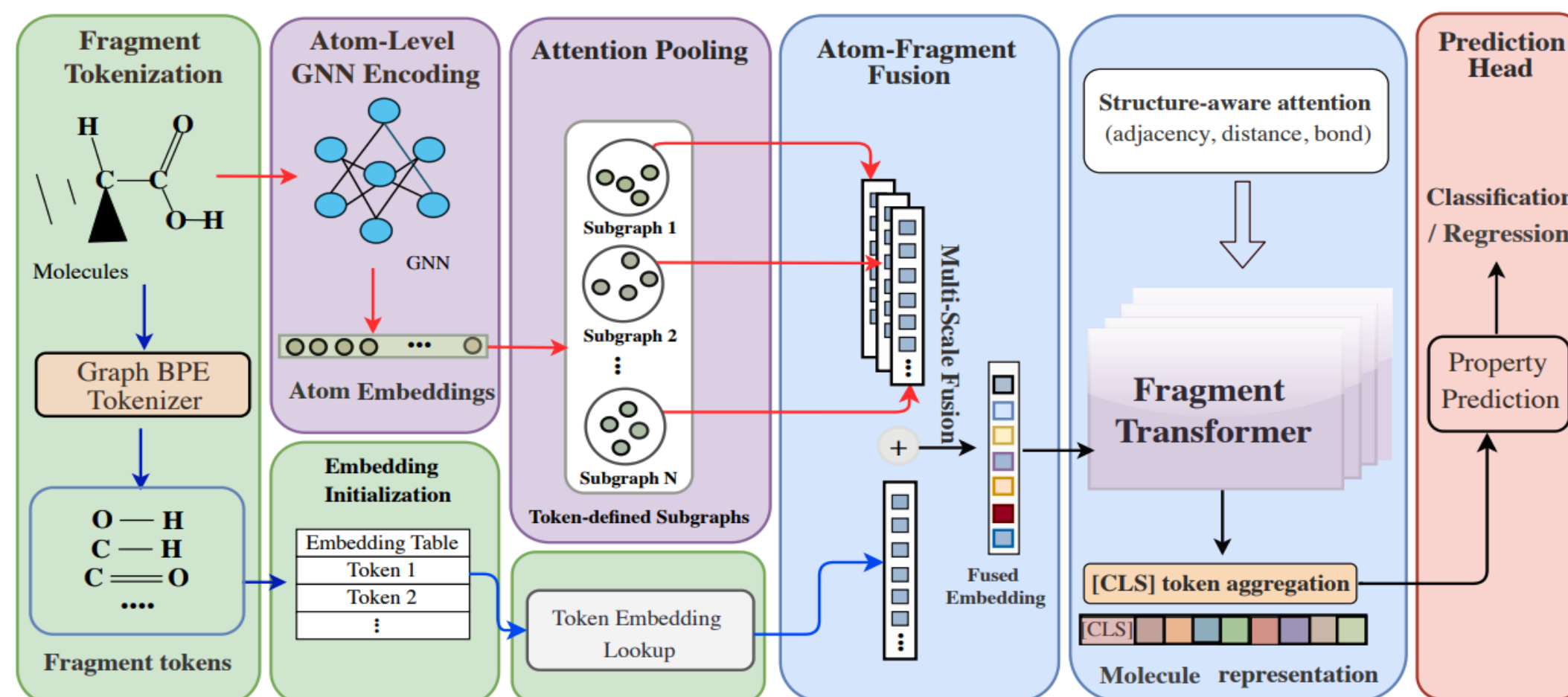
Challenges in Molecular Representation

- GNNs capture local structure well but suffer from over-smoothing and over-squashing for long-range dependencies.
- Transformers model global interactions via attention but lack strong structural inductive biases.

→ We need multi-scale representations

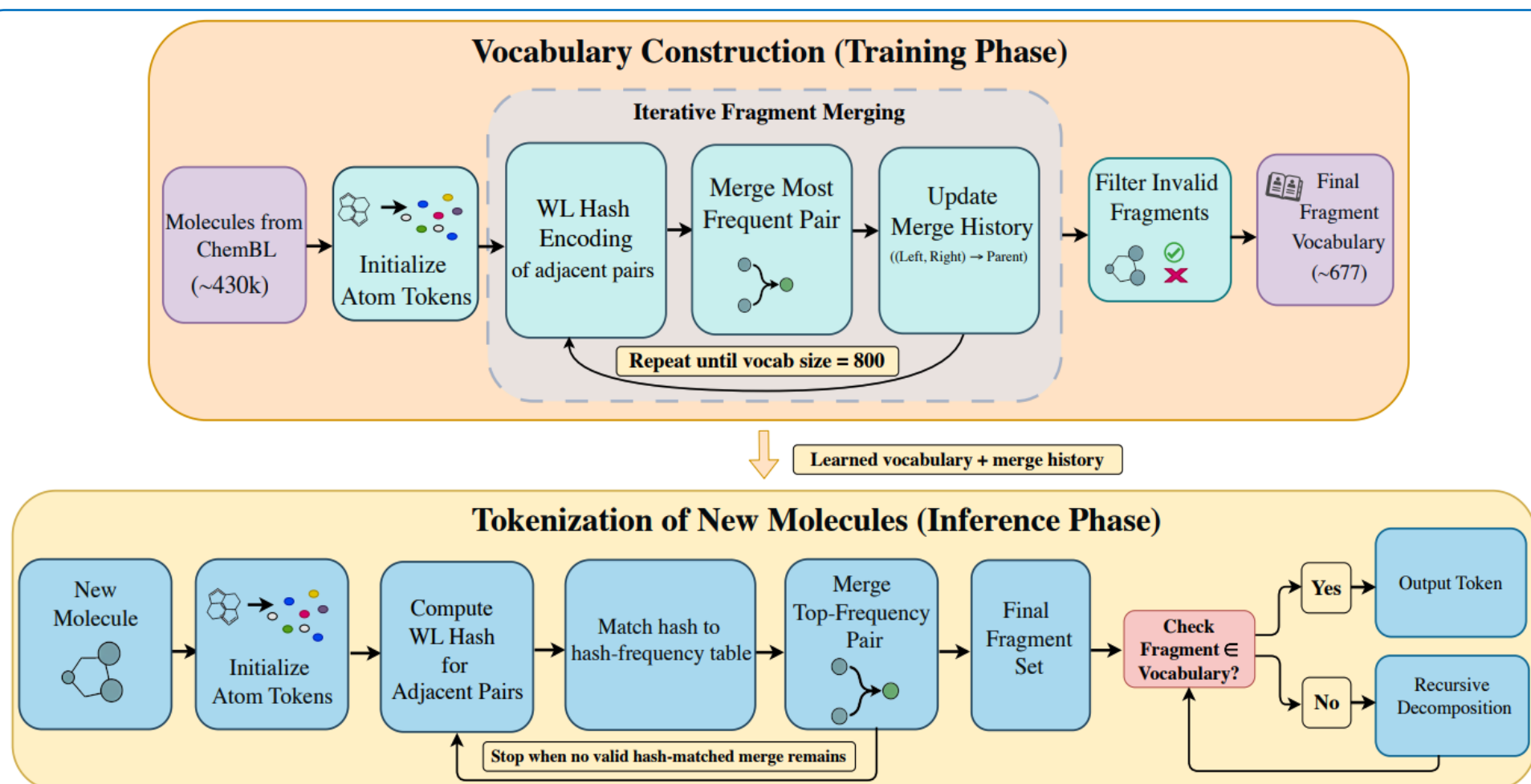


2. BiScale-GTR Framework Overview



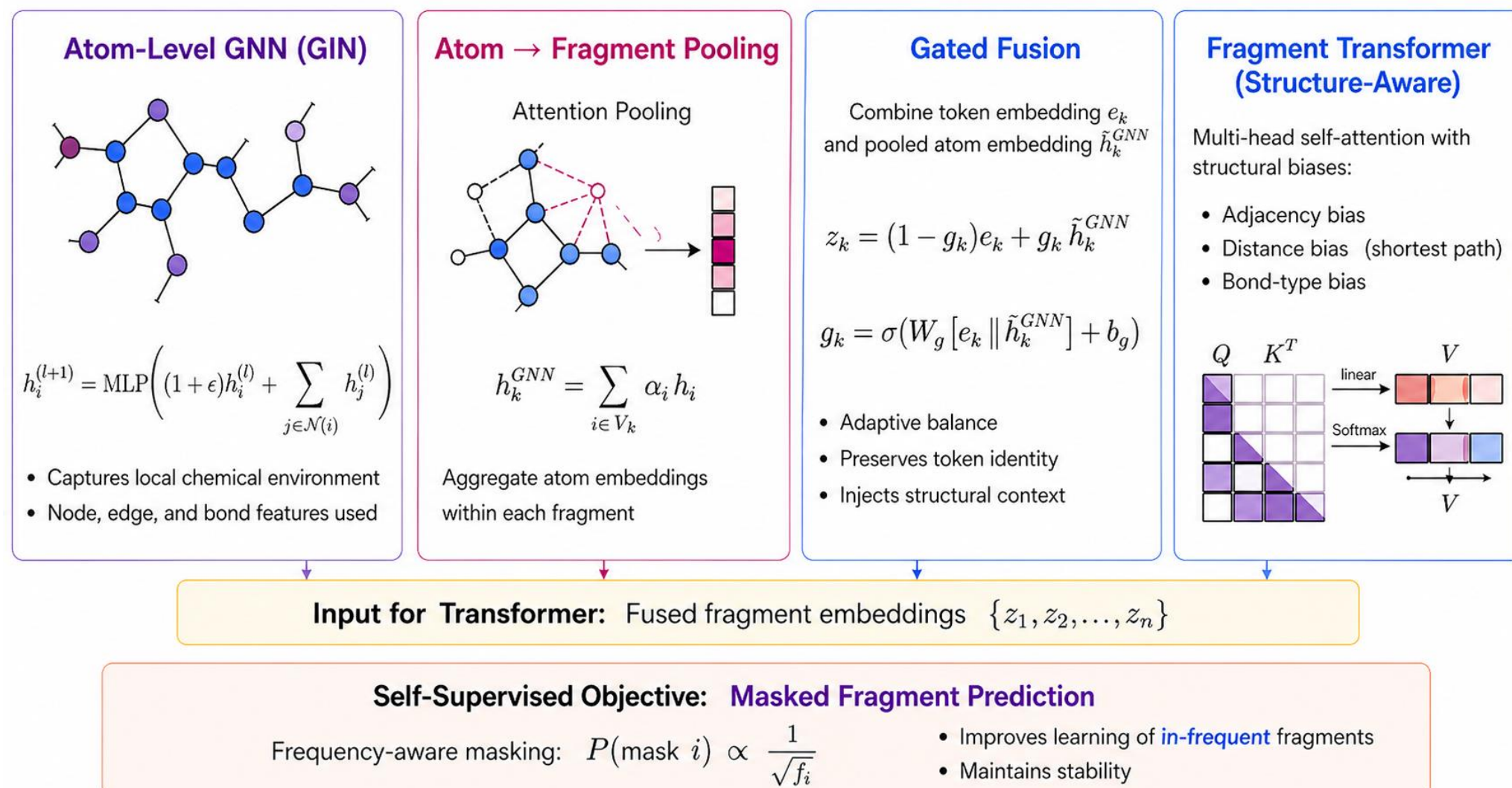
BiScale-GTR integrates atom-level structural encoding with fragment-level token representations. A fragment Transformer reasons over fused embeddings to capture both local and long-range dependencies.

3. Graph BPE Tokenizer



- Consistent via WL hashing
- Chemically valid through post generation filtering
- Recursive fallback for Out of vocabulary fragments

4. Model Architecture Details



5. Experimental Results

A. MoleculeNet (Classification, ROC-AUC)

Table 1: ROC-AUC (%) comparison on MoleculeNet biological classification tasks. Results are reported as mean $\pm$ standard deviation when available. Baseline results are taken directly from the corresponding original papers using the same data split protocol. **Bold** indicates the best result and underlined values denote the second-best performance.

Model	Pretrain Data Size	BBBP	Tox21	MUV	BACE	ToxCast	SIDER	HIV
GraphMVP (Liu et al., 2022)	50k	70.8 $\pm$ 0.5	74.9 $\pm$ 0.8	77.7 $\pm$ 0.6	79.3 $\pm$ 1.5	63.1 $\pm$ 0.2	60.2 $\pm$ 1.1	76.0 $\pm$ 0.1
GraphMAE (Hou et al., 2022)	2M	71.2 $\pm$ 1.0	75.2 $\pm$ 0.9	76.4 $\pm$ 2.0	78.2 $\pm$ 1.5	63.6 $\pm$ 0.3	60.5 $\pm$ 1.2	76.8 $\pm$ 0.6
Mole-BERT (Xia et al., 2023)	2M	71.9 $\pm$ 0.8	76.8$\pm$0.5	78.9 $\pm$ 1.8	80.8 $\pm$ 1.4	64.3 $\pm$ 0.2	62.8 $\pm$ 1.1	78.2 $\pm$ 0.8
SimSGT (Liu et al., 2023)	2M	72.2 $\pm$ 0.9	76.8 $\pm$ 0.9	81.4 $\pm$ 1.4	84.3 $\pm$ 0.6	65.9$\pm$0.8	61.7 $\pm$ 0.8	78.0 $\pm$ 1.9
GraphFP (Luong & Singh, 2023)	430k	72.0 $\pm$ 1.7	74.0 $\pm$ 0.7	75.4 $\pm$ 1.9	80.5 $\pm$ 1.8	63.9 $\pm$ 0.9	63.6 $\pm$ 1.2	78.0 $\pm$ 1.5
MORE (Son et al., 2025)	2M	71.9 $\pm$ 0.9	75.6 $\pm$ 0.5	–	82.8 $\pm$ 1.3	64.6 $\pm$ 0.6	60.9 $\pm$ 0.6	77.0 $\pm$ 0.7
GraphGPS-LAC (Yang et al., 2025)	–	73.6	74.0	71.3	82.5	73.7	60.4	77.6
BiScale-GTR (Fragment)	430k	73.8$\pm$0.4	73.1 $\pm$ 0.9	74.7 $\pm$ 0.4	83.8 $\pm$ 1.2	63.9 $\pm$ 0.2	60.1 $\pm$ 1.1	77.9 $\pm$ 1.1
BiScale-GTR (Molecule)	430k	68.4 $\pm$ 0.8	<u>76.1$\pm$0.4</u>	81.6$\pm$0.9	85.0$\pm$1.1	62.2 $\pm$ 0.2	64.2$\pm$0.9	79.2$\pm$0.6

B. PharmaBench (Regression, RMSE)

Table 2: Performance comparison on PharmaBench regression tasks (RMSE $\downarrow$). Baseline results are taken directly from the PharmaBench paper (Niu et al., 2024).

Model	CYP2C9	CYP2D6	CYP3A4	HLMC	MLMC	RLMC	LogD	PPB	Sol
MPG (Li et al., 2020)	17.417	17.527	14.376	0.541	0.723	0.685	0.758	0.170	0.758
UniMol (Zhou et al., 2023)	17.774	18.071	15.895	0.613	0.824	0.651	0.745	0.179	0.707
PharmHGT (Jiang et al., 2023)	17.490	15.020	16.077	0.544	0.820	0.677	0.676	0.172	0.954
FragFormer (Wang et al., 2025)	16.855	14.425	15.894	0.514	0.702	0.596	0.667	0.157	0.895
BiScale-GTR (Molecule)	16.633	16.901	16.011	0.501	0.696	0.571	0.801	0.153	0.977

Bold indicates the best (lowest RMSE).

C. LRGB (Classification and Regression)

Table 3: Performance comparison on peptide benchmarks. Average Precision (AP $\uparrow$) is reported for Peptides-func and Mean Absolute Error (MAE $\downarrow$) for Peptides-struct. Results are shown as mean $\pm$ standard deviation. Baseline results are taken directly from the LRGB benchmark.

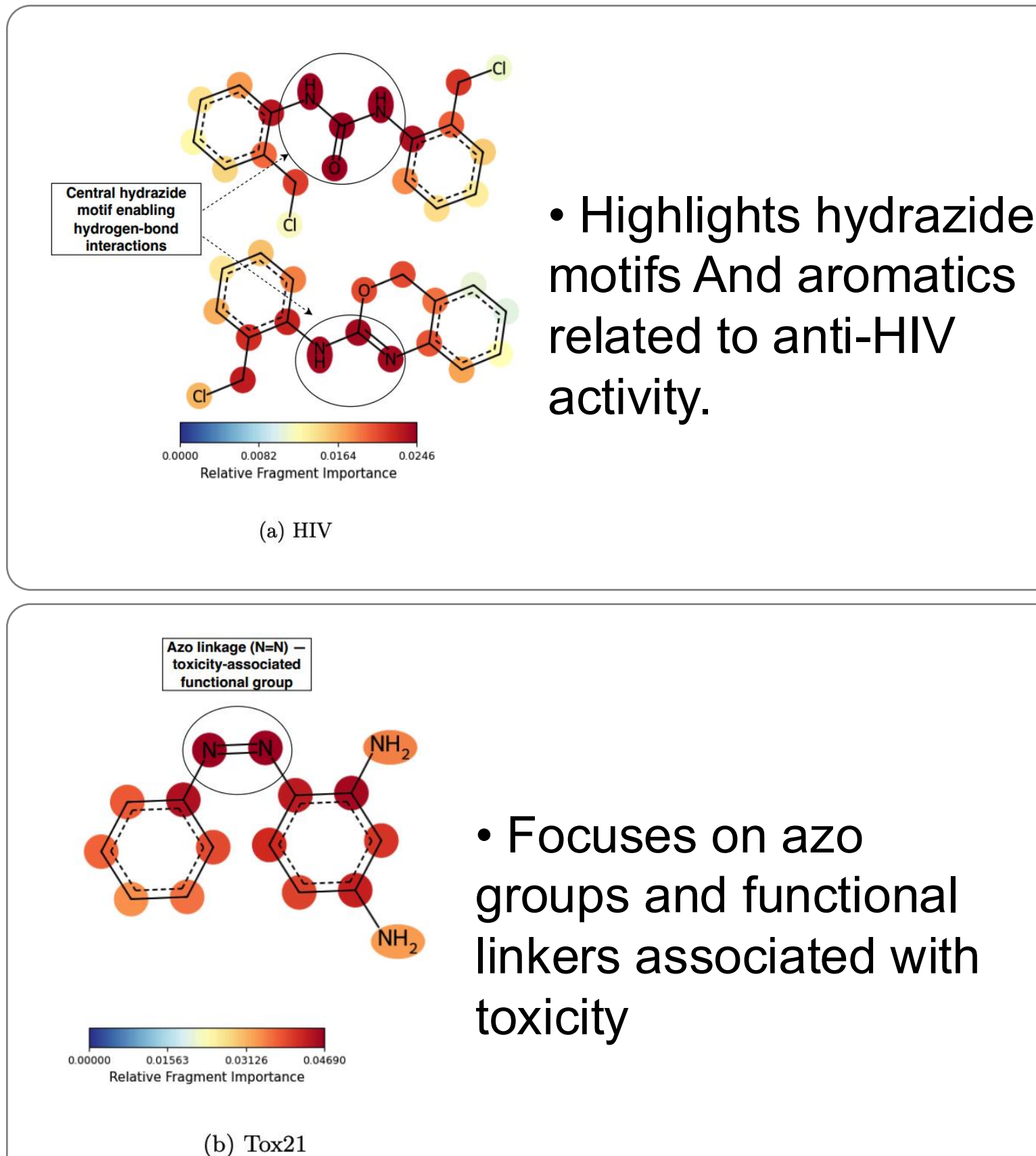
Model	Peptides-func (AP $\uparrow$)	Peptides-struct (MAE $\downarrow$)
GCN (Kipf & Welling, 2016)	0.5930 $\pm$ 0.0023	0.3496 $\pm$ 0.0013
GCNII (Chen et al., 2020)	0.5543 $\pm$ 0.0078	0.3471 $\pm$ 0.0010
GINE (Hu et al., 2019)	0.5498 $\pm$ 0.0079	0.3547 $\pm$ 0.0045
GatedGCN (Bresson & Laurent, 2017)	0.5864 $\pm$ 0.0077	0.3420 $\pm$ 0.0013
GatedGCN+RWSE (Dwivedi et al., 2022)	0.6069 $\pm$ 0.0035	0.3357 $\pm$ 0.0006
Transformer (Vaswani et al., 2017) + LapPE (Dwivedi et al., 2023)	0.6326 $\pm$ 0.0126	0.2529$\pm$0.0016
SAN (Kreuzer et al., 2021) + LapPE	0.6384 $\pm$ 0.0121	0.2683 $\pm$ 0.0043
SAN + RWSE	0.6439 $\pm$ 0.0075	0.2545 $\pm$ 0.0012
GraphFP (Luong & Singh, 2023)	0.6267 $\pm$ 0.0073	0.3137 $\pm$ 0.0019
FragFormer (Wang et al., 2025)	0.6693 $\pm$ 0.0154	–
BiScale-GTR (Molecule)	0.6717$\pm$0.0107	0.2621 $\pm$ 0.0022



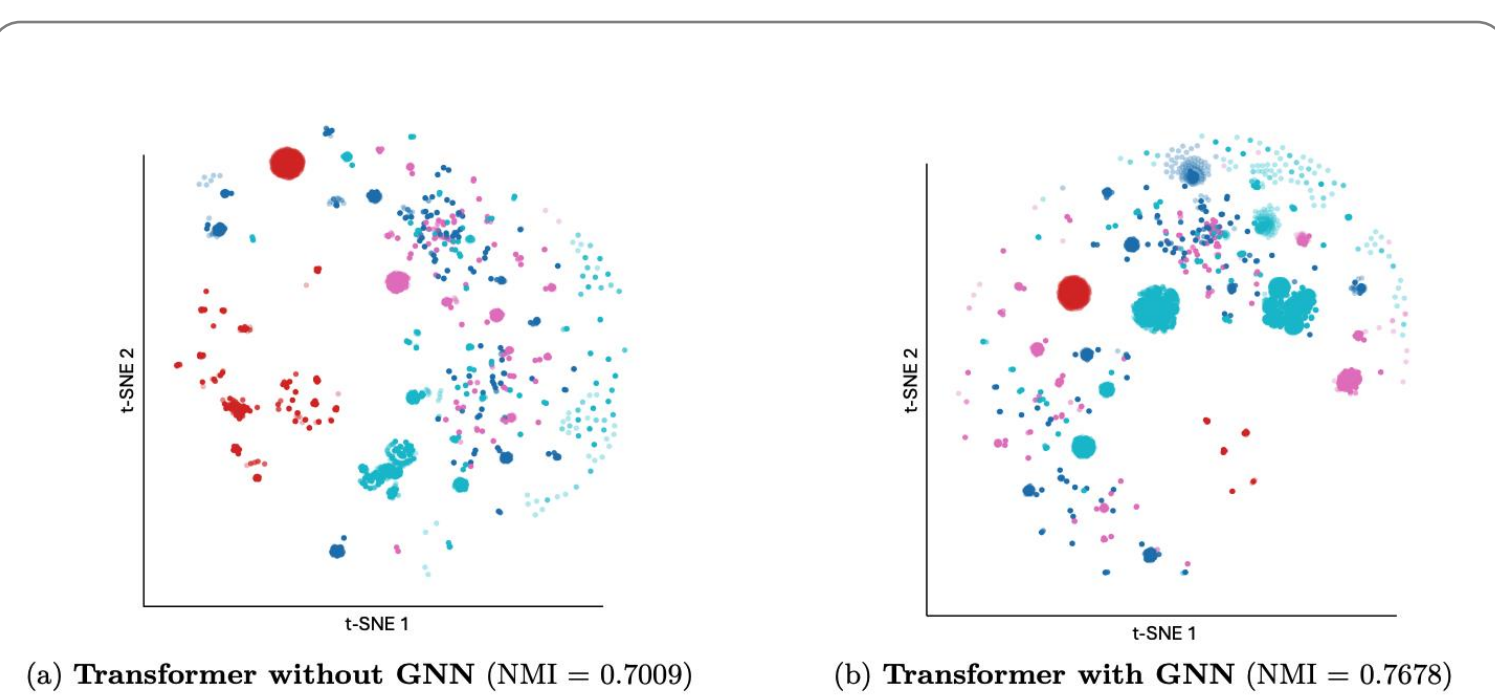
BiScale-GTR achieves state-of-the-art performance across multiple benchmarks in both classification and regression.

6. Interpretability & Analysis

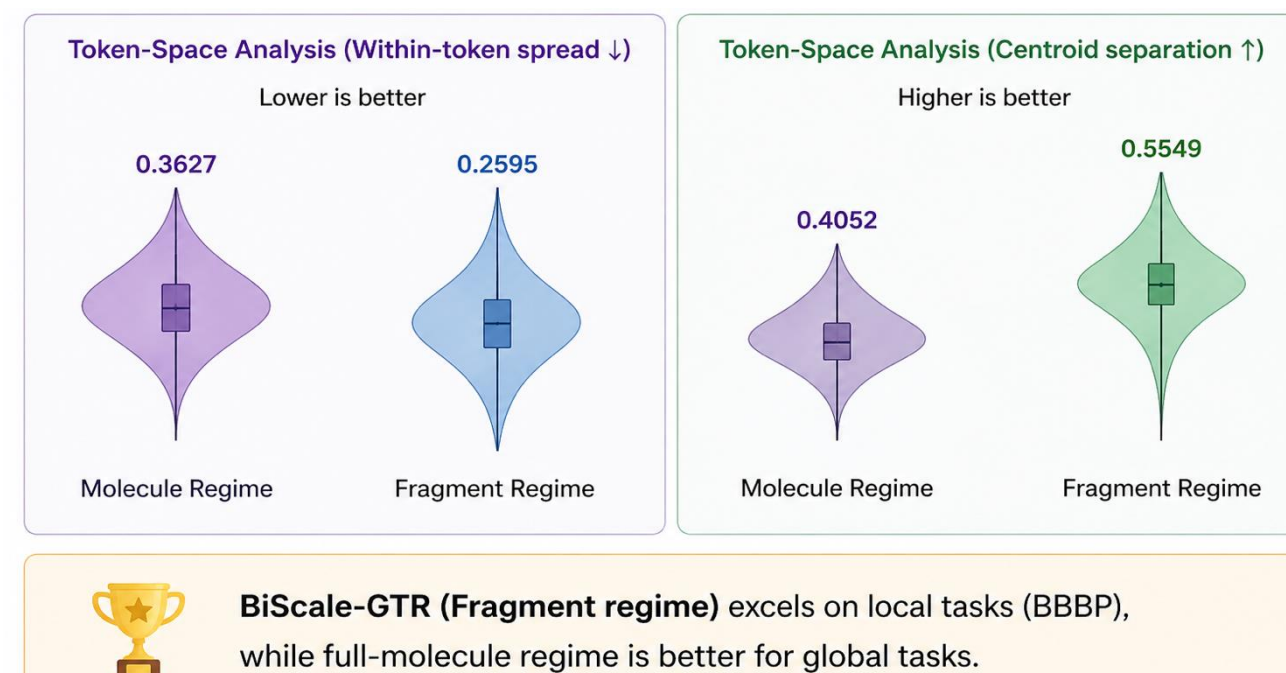
A. Attention Attribution (Fragment Level)



B. Fragment Embedding Visualization (t-SNE)



C. Two-Regime Analysis (BBBP)



7. Contributions

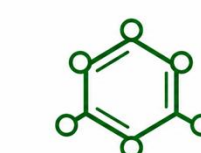
- ✓ We propose BiScale-GTR, a novel multi-scale framework that integrates atom-level GNN encoding with fragment-level Transformer reasoning
- ✓ We design a consistent and chemically valid Graph BPE tokenizer for fragment discovery and tokenization.
- ✓ We provide extensive analysis showing interpretability, embedding quality, and task-dependent advantages.
- ✓ We achieve state-of-the-art performance on MoleculeNet, PharmaBench, and LRGB benchmarks.
- ✓ We introduce an adaptive gated fusion mechanism to combine token identity and structural context

8. Datasets

- MoleculeNet: 7 classification datasets
- PharmaBench: 9 ADMET regression datasets
- LRGB: long-range graph property prediction

9. Takeaway

BiScale-GTR bridges local chemistry and global structure through fragment-aware multi-scale learning.



Chemically grounded fragments



Multi-scale representation



Better performance & interpretability

Scan for the paper !



SCAN ME



GitHub Repository

<https://github.com/AI4Science2025>



Code will be released

The source code and related resources will be released when the paper is accepted.

